

FUNDATA, Romania

WMO: 153010

Lat: 45.4314N

Lon: 25.2717E

Elev: 4546

StdP: 12.44

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	7.4	-2.0	5.9	10.3	2.3	7.3	12.1	31.8	18.4	26.5	18.6	8.9	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	74.0	62.9	71.5	60.9	69.2	59.8	65.0	72.0	62.9	69.1	61.2	67.2	5.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.5	100.3	69.3	60.5	93.3	66.7	58.8	87.6	64.8	32.6	72.0	31.0	69.0	29.6	67.5	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	17.3	14.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	23.1	28.1	31.8	40.1	49.8	57.4	59.0	60.3	52.2	43.4	35.8	27.6		
	DBStd	15.11	9.59	8.98	8.62	9.92	6.85	5.85	5.43	6.04	8.48	9.19	9.14	8.47		
	HDD50	3886	834	615	566	319	84	10	3	3	79	249	431	693		
	HDD65	8268	1299	1035	1029	746	471	235	195	165	389	669	877	1158		
	CDD50	1134	0	0	1	24	78	231	283	324	145	44	4	0		
	CDD65	42	0	0	0	0	0	5	10	21	6	0	0	0		
	CDH74	72	0	0	0	0	0	6	19	33	14	0	0	0		
	CDH80	3	0	0	0	0	0	0	0	1	2	0	0	0		
(14) Wind	WSAvg	6.5	7.7	7.6	7.2	6.7	6.1	5.6	5.4	5.1	5.7	6.3	7.5	7.4		
(15) Precipitation	PrecAvg	34.8	1.8	1.7	2.0	2.8	4.1	4.7	4.4	3.8	2.9	2.5	2.1	2.0		
	PrecMax	48.8	4.9	3.3	3.7	5.7	7.7	8.5	7.8	8.1	5.8	5.4	4.2	4.8		
	PrecMin	22.6	0.3	0.3	0.5	1.4	1.4	2.0	0.5	1.2	0.8	0.2	0.2	0.3		
	PrecStd	6.5	1.1	0.7	0.9	1.1	1.6	1.8	1.9	1.8	1.4	1.3	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	50.8	56.3	64.9	69.4	74.6	76.3	77.9	75.0	67.4	59.2	48.9		
		MCWB	37.0	42.0	43.8	51.3	56.3	61.1	64.9	65.0	64.9	52.1	47.6	37.4		
	2%	DB	41.6	46.6	51.6	61.4	66.1	71.1	72.9	74.2	69.6	63.1	54.3	45.1		
		MCWB	35.9	38.4	40.8	49.5	54.9	60.3	62.3	63.3	58.4	51.0	45.2	37.1		
	5%	DB	38.8	43.0	47.4	57.8	62.9	68.7	70.2	71.9	66.3	59.5	50.8	41.9		
		MCWB	34.8	36.2	38.9	47.5	53.7	59.3	61.2	61.5	57.0	49.8	43.7	36.8		
	10%	DB	35.9	40.0	43.7	54.0	59.9	66.3	67.6	69.3	63.4	55.9	48.0	38.6		
		MCWB	32.4	35.0	37.2	45.4	52.3	58.3	59.9	60.4	55.1	47.5	42.8	35.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	44.4	44.7	52.5	58.9	63.9	66.5	67.7	65.3	55.6	49.2	42.1		
		MCDB	43.1	49.3	53.8	62.0	64.7	71.2	73.8	73.0	73.4	61.0	55.4	44.0		
	2%	WB	37.5	40.3	41.8	50.5	56.4	61.9	64.6	65.3	61.0	52.4	46.7	38.7		
		MCDB	40.3	44.8	49.5	59.8	63.1	68.2	70.7	72.2	66.8	60.1	52.5	42.4		
	5%	WB	35.6	37.5	39.6	48.6	54.6	60.3	62.4	63.4	58.0	50.6	44.8	37.2		
		MCDB	38.3	42.0	46.5	56.4	61.6	66.4	67.5	70.3	64.0	58.4	49.8	41.0		
	10%	WB	32.7	35.6	37.6	46.5	52.7	59.0	60.9	61.2	55.9	48.6	43.4	35.1		
		MCDB	35.4	38.7	43.1	53.0	59.0	65.5	66.3	67.7	62.1	54.9	47.6	38.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.5	11.1	12.8	12.3	12.7	12.4	12.2	12.0	11.3	9.9	9.3		
		MCDBR	12.4	13.7	15.5	16.1	15.0	14.3	14.4	13.9	14.4	14.7	12.4	11.9		
	5% WB	MCWBR	8.4	8.8	9.0	9.0	9.2	8.9	8.8	8.9	8.1	8.6	7.4	8.0		
		MCWBR	12.4	12.1	14.3	14.7	14.8	13.7	13.3	13.3	12.8	13.9	10.9	10.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.293	0.329	0.379	0.363	0.364	0.376	0.388	0.351	0.320	0.299	0.271		
		taud	2.381	2.326	2.318	2.244	2.340	2.357	2.323	2.287	2.379	2.466	2.479	2.430		
	Ebn at Noon	Ebn at Noon	270	282	285	279	287	286	281	272	275	270	257	257		
		Edh at Noon	23	30	35	41	39	38	39	39	33	26	21	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	444	705	1045	1362	1659	1796	1827	1652	1211	837	489	363		
	RadStd	RadStd	46	90	109	150	162	162	156	137	117	107	50	44		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (74 neighbors)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.06	N/A	N/A	+1.31	N/A	N/A	-230	-313	+137	+43

Nomenclature: See separate page